Final ML Project Report: Stroke Prediction Classification

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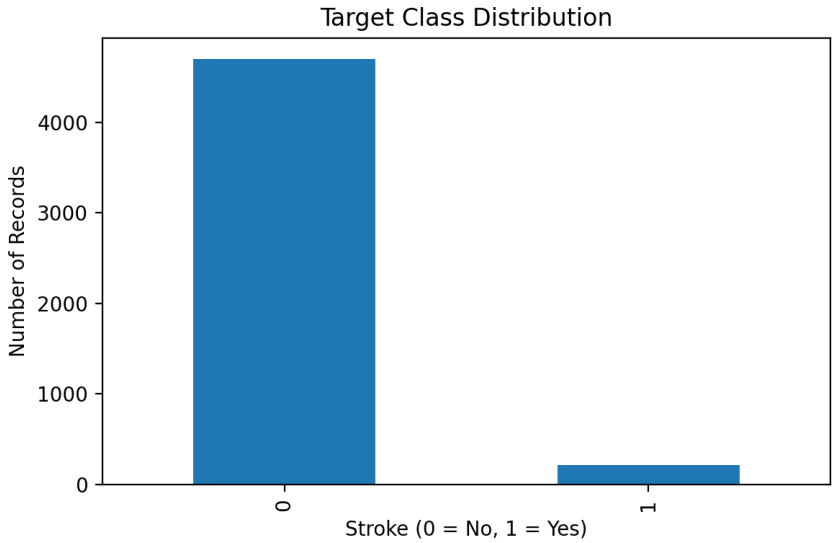
Project Type: Classification
Dataset: healthcare-dataset-stroke-cleaned.csv

1. Project Goal

The goal of this project is to build and compare machine learning classification models that predict whether a patient has experienced a stroke using demographic and health-related features. The target variable is **stroke**, where 0 means no stroke and 1 means stroke.

2. Dataset and Preprocessing

The cleaned dataset contains 4909 records and 11 columns. There are 4700 non-stroke records and 209 stroke records, so the data is highly imbalanced. Because the preprocessing was already completed, this project used the cleaned features directly. The remaining categorical feature, age_group, was one-hot encoded, and numeric features were scaled inside each model pipeline to avoid data leakage.



3. Train, Validation, and Test Split

The dataset was split using stratification so the same class ratio was preserved across the sets. The training set has 3436 records (70%), the validation set has 736 records (15%), and the test set has 737 records (15%). Model selection was based on validation performance, while final performance was checked on the unseen test set.

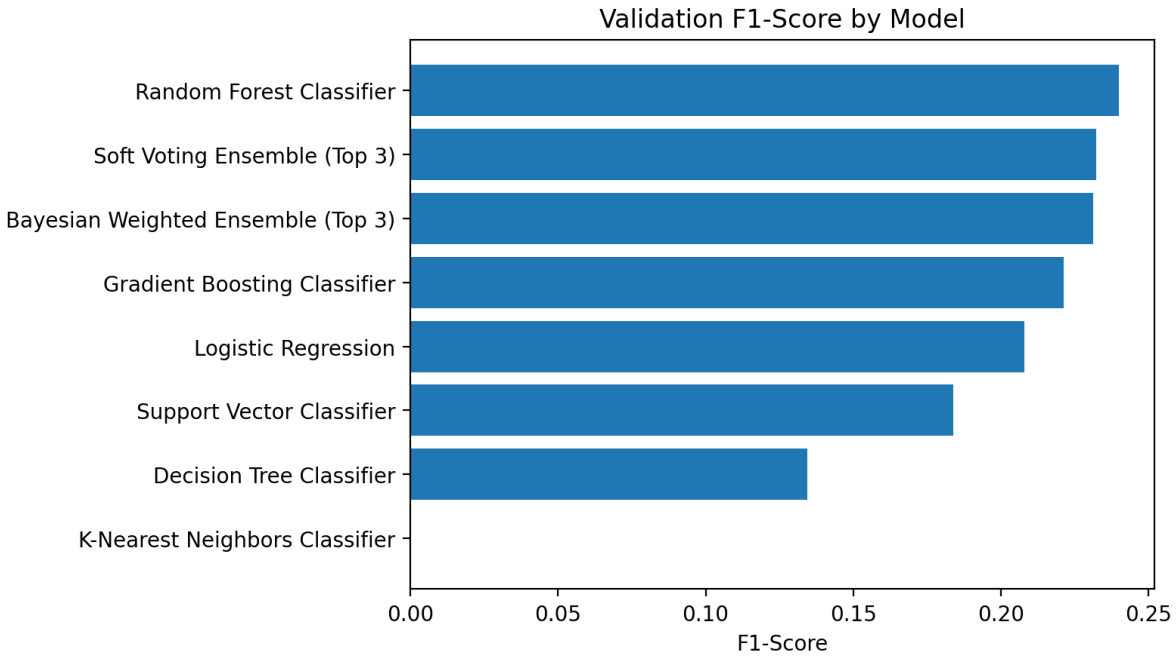
4. Models Trained

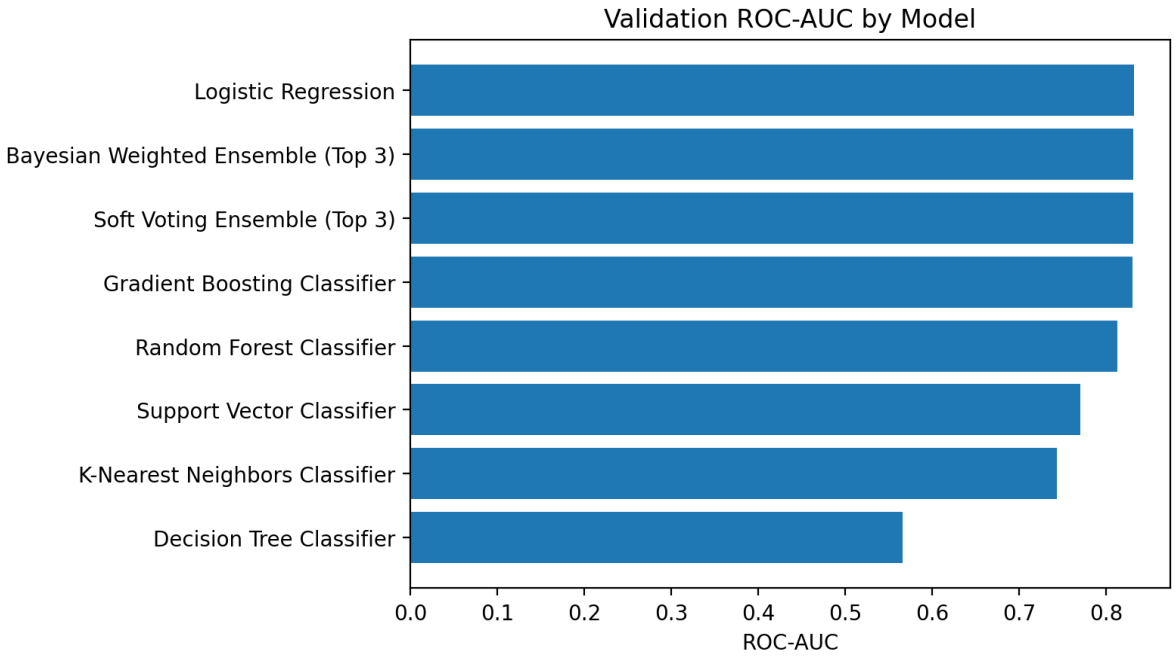
The required models were trained: Logistic Regression, Decision Tree Classifier, Random Forest Classifier, Gradient Boosting Classifier, K-Nearest Neighbors Classifier, and Support Vector Classifier. Class weighting or sample weighting was used where possible because stroke cases are much less common than non-stroke cases.

5. Evaluation Metrics

The models were compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Because this is a medical classification problem with imbalanced classes, F1-Score and ROC-AUC are more useful than

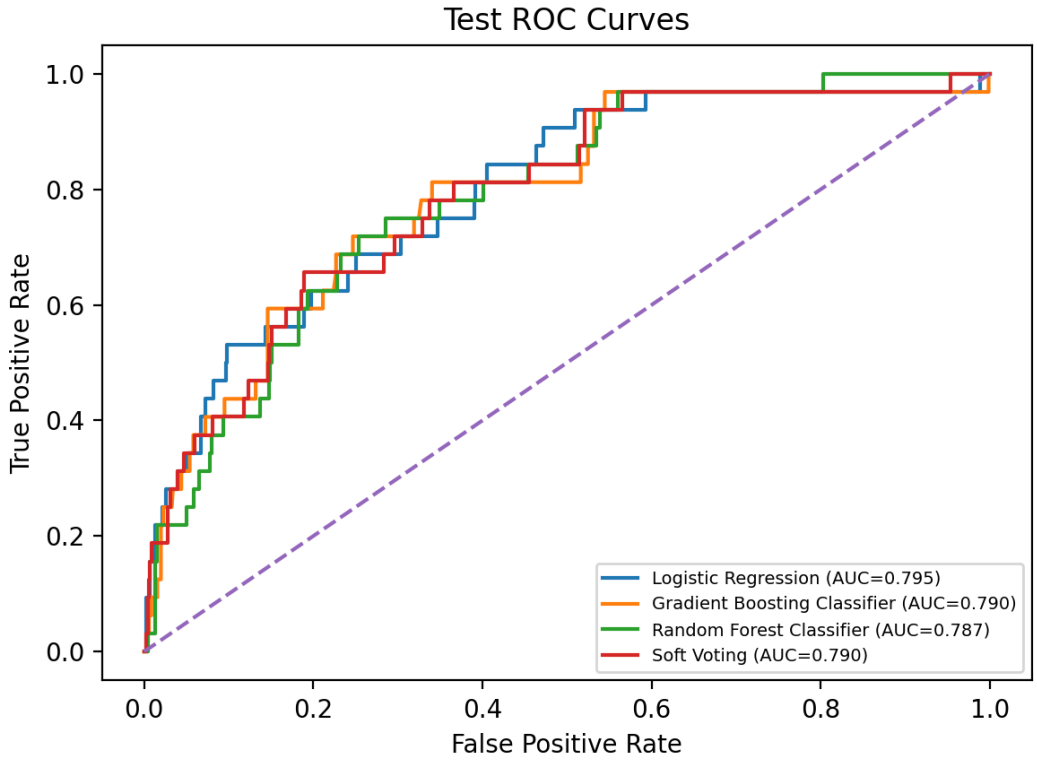
accuracy alone. A model can have high accuracy by predicting mostly no-stroke cases, but that does not mean it is good at finding positive stroke cases.





6. Ensemble Models

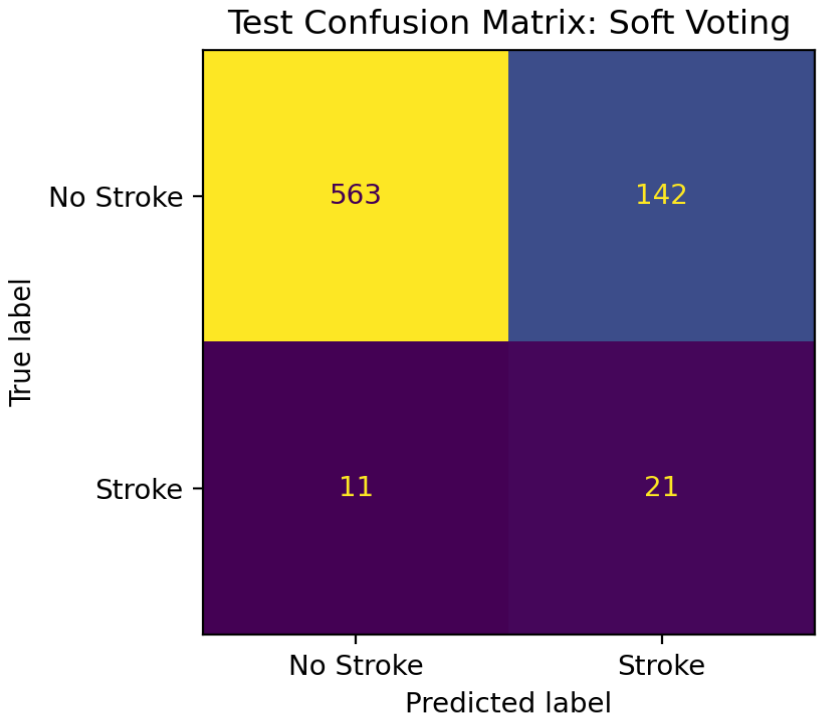
The top three models by validation ROC-AUC were Logistic Regression, Gradient Boosting Classifier, and Random Forest Classifier. A soft voting ensemble averaged their predicted probabilities. A Bayesian weighted ensemble used validation ROC-AUC scores as model evidence and normalized them into weights: Logistic Regression=0.336, Gradient Boosting Classifier=0.335, Random Forest Classifier=0.328. This lets stronger validation models contribute slightly more to the final probability.



7. Best Model and Discussion

On the validation set, the strongest individual model by F1-Score was Random Forest Classifier with F1=0.2400. On the test set, the strongest overall approach by F1-Score was Soft Voting Ensemble (Top 3) with F1=0.2154, Recall=0.6562, and ROC-AUC=0.7901. The ensemble performed well because it combined the probability outputs from the strongest models instead of relying on one model only.

The K-Nearest Neighbors model had high accuracy but failed to identify stroke cases at the default threshold. This shows why accuracy alone is not enough for imbalanced healthcare data. Logistic Regression achieved strong recall and ROC-AUC, Gradient Boosting gave a good balance of recall and probability ranking, and Random Forest had the best validation F1 among individual models. The ensemble gave the best test F1 by combining these strengths.



8. Conclusion

The final recommendation is to use the soft voting ensemble or Bayesian weighted ensemble because they produced the strongest test F1-Score while maintaining useful recall for stroke cases. For a real healthcare use case, future work should include threshold tuning with clinical input, stronger imbalance handling, feature importance review, and validation on a larger external dataset.